Fractal Interpolation :

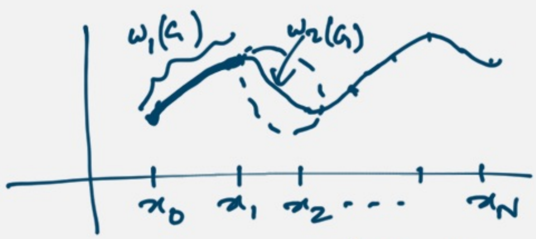
given : $\{(x_i, F_i) : i = 0, 1, 2, \ldots, N\}$ a set of data

We want a continuous fn. $f : [x_0, x_N] \to \mathbb{R}$ which interpolates the data, ie., $f(x_i) = F_i$ for $i = 0, 1, \ldots, N$, such that the graph of f is an attractor of an IFS on $\mathbb{R}^2$.

We consider an IFS of the form $\{\mathbb{R}^2; w_1, w_2, \ldots, w_N\}$ where the maps are affine linear maps of the form

$$w_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & d_n \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix}$$

We want : $G = \bigcup_{n=1}^{N} w_n(G)$, where $G = \text{graph}(f)$.



Assume $w_n \begin{pmatrix} x_0 \\ F_0 \end{pmatrix} = \begin{pmatrix} x_{n-1} \\ F_{n-1} \end{pmatrix}$ and $w_n \begin{pmatrix} x_N \\ F_N \end{pmatrix} = \begin{pmatrix} x_n \\ F_n \end{pmatrix}$ for $n = 1, 2, \ldots, N$.

This gives
$$\begin{cases} a_n x_0 + e_n = x_{n-1} \\ a_n x_N + e_n = x_n \\ c_n x_0 + d_n F_0 + f_n = F_{n-1} \\ c_n x_N + d_n F_N + f_n = F_n \end{cases}$$

There are four eqns. in five unknowns a_n, c_n, d_n, e_n, f_n

Let's choose d_n as the free parameter.

Then

$$(*)\begin{cases} a_n = \dfrac{x_n - x_{n-1}}{x_N - x_0} \\ e_n = \dfrac{x_N x_{n-1} - x_0 x_N}{x_N - x_0} \\ c_n = \dfrac{(F_n - F_{n-1})}{(x_N - x_0)} - d_n \dfrac{(F_n - F_0)}{(x_N - x_0)} \\ f_n = \left(\dfrac{x_N F_{n-1} - x_0 F_n}{x_N - x_0}\right) - d_n \left(\dfrac{x_N F_0 - x_0 F_N}{x_N - x_0}\right). \end{cases}$$

- d_n is called the "vertical scaling factor".

  We choose $0 \leq d_n < 1$.

Theorem: Let $N > 1$ and let $\{\mathbb{R}^2; w_1, w_2, \dots, w_N\}$ denote the IFS defined above, associated with the data set $\{(x_i, F_i) : i = 0, 1, \dots, N\}$.
Let $0 \leq d_n < 1$ for $n = 1, 2, \dots, N$.
Then there is a metric d on $\mathbb{R}^2$, which is equivalent to the Euclidean metric, such that the IFS is hyperbolic with respect to d (i.e. the maps w_n's are contractions w.r.t. d).
In particular, $\exists$ a unique nonempty compact set $G \subseteq \mathbb{R}^2$ s.t. $G = \bigcup_{n=1}^{N} w_n(G)$.

Proof: We define a metric d on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + \theta |y_1 - y_2|,$$

where $\theta > 0$ is a real no. which we specify below.

It can be easily shown that this metric d is equivalent to the Euclidean metric. (Exercise)

Let a_n, c_n, e_n, f_n be as defined in $(*)$.

Then $d(w_n(x_1,y_1), w_n(x_2,y_2))$

$$= d\left((a_n x_1 + e_n, c_n x_1 + d_n y_1 + f_n), (a_n x_2 + e_n, c_n x_2 + d_n y_2 + f_n)\right)$$

$$= |a_n|\,|x_1 - x_2| + \theta\,|c_n(x_1-x_2) + d_n(y_1-y_2)|$$

$$\leq (|a_n| + \theta|c_n|)\,|x_1-x_2| + \theta\,|d_n|\,\underline{|y_1-y_2|} \quad (i)$$

Note that $a_n = \dfrac{x_n - x_{n-1}}{x_N - x_0} \in (0,1)$ $\forall n = 1,2,\dots,N$.

(because $N \geq 2$).

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From (i),

$d(w_n(x_1,y_1), w_n(x_2,y_2))$

$$\leq \max\{a_n + \theta|c_n|, |d_n|\}\, d((x_1,y_1),(x_2,y_2))$$

We choose $\theta > 0$ s.t. $a_n + \theta|c_n| < 1$ $\forall n$

i.e. $\theta|c_n| < 1 - a_n$

If $c_n = 0$ $\forall n = 1,2,\dots,N$, we choose $\theta = 1$

If $c_n \neq 0$ for some n, then we take

$$\theta = \frac{\min\{1 - a_n : n = 1,2,\dots,N\}}{2\max\{|c_n| : n = 1,2,\dots,N\}}$$

This gives $a_n + \theta|c_n| \leq \frac{1}{2}$ $\forall n$.

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Then each w_n is a contraction map w.r.t. the metric d.

Theorem 2. Let $N \geqslant 2$ and let $\{\mathbb{R}^2, w_1, w_2, \dots, w_N\}$ denote the IFS defined above, associated with the data set $\{(x_i, F_i) : i = 0, 1, \dots, N\}$. Let the vertical scaling factor $d_n \in [0,1)$ for $n = 1, 2, \dots, N$, so that IFS is hyperbolic. Let G denote the attractor of this IFS. Then G is the graph of a continuous function $f : [x_0, x_N] \to \mathbb{R}$ which interpolates the given data, i.e.,

$$G = \{(x, f(x)) : x \in [x_0, x_N]\}$$

and $f(x_i) = F_i$ for $i = 0, 1, 2, \dots, N$.

Proof: Let $\mathcal{F} = \{ f \in C([x_0, x_N], \mathbb{R}) : f(x_0) = F_0 \text{ \& } f(x_N) = F_N \}$

Consider the metric d on $\mathcal{F}$ given by

$$d(f, g) = \max\{ |f(x) - g(x)| : x \in [x_0, x_N] \}$$

Then it is easy to show that $(\mathcal{F}, d)$ is a complete metric space.

Define $T : \mathcal{F} \to \mathcal{F}$ by

$$T(f)(x) = c_n l_n^{-1}(x) + d_n f(l_n^{-1}(x)) + f_n$$

for $x \in [x_{n-1}, x_n]$, $n \in \{0, 1, 2, \dots, N\}$,

where $l_n : [x_0, x_N] \to [x_{n-1}, x_n]$ is the map

$$l_n(x) = a_n x + e_n$$

Verify that T is a well-defined map.